

CATHOLIC UNIVERSITY OF GHANA, FIAPRE – SUNYANI

FACULTY OF CEMS



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INDEX: UGR0202210065

COURSE CODE: SCSC 452

**COURSE TITLE: WIRELESS TECHNOLOGIES AND MOBILE
COMPUTING**

PROGRAM: BSC INFORMATION TECHNOLOGY

LEVEL: 400

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DATE: 4TH FEBRUARY 2026

ASSIGNMENT: 1

1. Differentiate between Piconet and Scatternet.

A **piconet** is the basic network structure in Bluetooth technology. It consists of **one master device** and up to **seven active slave devices** that communicate with the master using Bluetooth.

The master controls all communication in a piconet, and slaves cannot communicate directly with each other. The devices within a piconet operate in a short range, typically up to **10 meters**, making it suitable for personal device connectivity such as phones, headsets, and laptops, but A **scatternet**, on the other hand, is formed when **two or more piconets are interconnected**. In a scatternet, a device can act as a **master in one piconet and a slave in another**, allowing data to be forwarded between piconets. This enables wider coverage and supports more devices compared to a single piconet. Thus, while a piconet is a simple Bluetooth network, a scatternet is a more complex structure that extends Bluetooth networking capabilities.

2. Differentiate between Basic Service Set (BSS) and Extended Service Set (ESS).

A **Basic Service Set (BSS)** is the fundamental building block of an IEEE 802.11 wireless LAN.

It consists of a group of wireless stations that communicate with each other at the physical layer.

A BSS can operate in **infrastructure mode**, where all stations communicate through an Access Point (AP), or in **independent mode (IBSS)**, where stations communicate directly with each

other in an ad-hoc manner without an AP but An **Extended Service Set (ESS)** is formed by

connecting multiple BSSs through a **Distribution System (DS)**, which may be wired or wireless. The ESS allows users to move from one BSS to another while maintaining continuous network connectivity. To external networks, an ESS appears as a **single large WLAN**, even though it consists of multiple access points and coverage areas. Hence, BSS provides local connectivity, whereas ESS provides larger coverage and seamless mobility.

3. Explain the concept of IEEE 802.11 architecture.

The **IEEE 802.11 architecture** defines the structure and operation of Wireless Local Area Networks (WLANs). It is designed to support wireless communication among fixed, portable, and mobile devices within a limited area using radio waves. The architecture focuses on the **Physical (PHY) layer** and the **Medium Access Control (MAC) sublayer** of the OSI model.

The main components of this architecture include **stations (STAs)** such as laptops and smartphones, **Access Points (APs)** that act as base stations, the **Basic Service Set (BSS)** for local communication, the **Extended Service Set (ESS)** for broader network coverage, and the **Distribution System (DS)** that interconnects multiple access points. IEEE 802.11 supports both **infrastructure networks** (with APs) and **ad-hoc networks** (without APs). The architecture is distributed in nature, which improves fault tolerance, flexibility, and power efficiency.

4. Explain the three major issues in the wireless environment

One major issue in the wireless environment is **half-duplex operation**. In wireless communication, a device cannot transmit and receive data simultaneously because the transmitted signal is much stronger than the received signal. This makes collision detection difficult, so wireless protocols focus on **collision avoidance** instead of collision detection.

The second issue is the **time-varying channel**. Wireless signals are affected by reflection, diffraction, and scattering, causing multipath propagation. As a result, signal strength fluctuates over time, leading to fading. To ensure reliable communication, techniques such as **handshaking** are used to verify link quality before data transmission.

The third issue is **burst channel errors**. Due to fading and interference, wireless links experience high bit error rates, often resulting in packet losses occurring in bursts rather than

randomly. To mitigate this, methods such as **smaller packet sizes**, **forward error correction**, and **retransmissions** are employed.

5. Demonstrate with examples of Wireless Personal Area Network (WPAN) and Wireless Local Area Network (WLAN).

A **Wireless Personal Area Network (WPAN)** covers a very short range and is designed for personal device connectivity. A common example is **Bluetooth**, which is used to connect devices such as mobile phones, wireless headsets, keyboards, mice, and smartwatches. Bluetooth networks operate in the 2.4 GHz ISM band and form piconets and scatternets for short-distance communication.

A **Wireless Local Area Network (WLAN)** covers a larger area such as homes, offices, campuses, or public hotspots. An example is **Wi-Fi (IEEE 802.11)**, which connects laptops, smartphones, printers, and servers through access points to provide internet and local network access. WLANs support mobility within the coverage area and are widely used in educational institutions, offices, and public places.

Reference

[1] MOBILE COMPUTING B. Tech CSE